

NRLC-09: Toolchain Reference & End-to-End Runbook

Series: NRLC | Notebook: 9 of 9 | Created: April 2026 | Last Updated: 04/17/2026

Overview

The master reference for the open-source toolchain that powers every preceding notebook in this series. This deep dive covers how `Dynatrace-NewRelic`, `nrql-engine`, and `nrql-translator` fit together, how to operate the orchestrating CLI, the configuration surface, the export formats (Monaco v2 YAML, Terraform HCL), and a complete end-to-end runbook.

As of 2026-04-15 the Python orchestrator ships **40+ transformers** across 9 phase batches (Phase 11 foundation + Phases 16–24 expansion), **1,200+ unit tests**, and **292 NRQL translation patterns** pinned to the TypeScript `nrql-engine` via the Phase 19b regression suite. It emits **Gen3 Dynatrace by default** (Workflows, anomaly detectors, Segments, OpenPipeline, Document API); use `--legacy` only for the 8 Gen2-only capabilities or for tenants without Gen3 features.

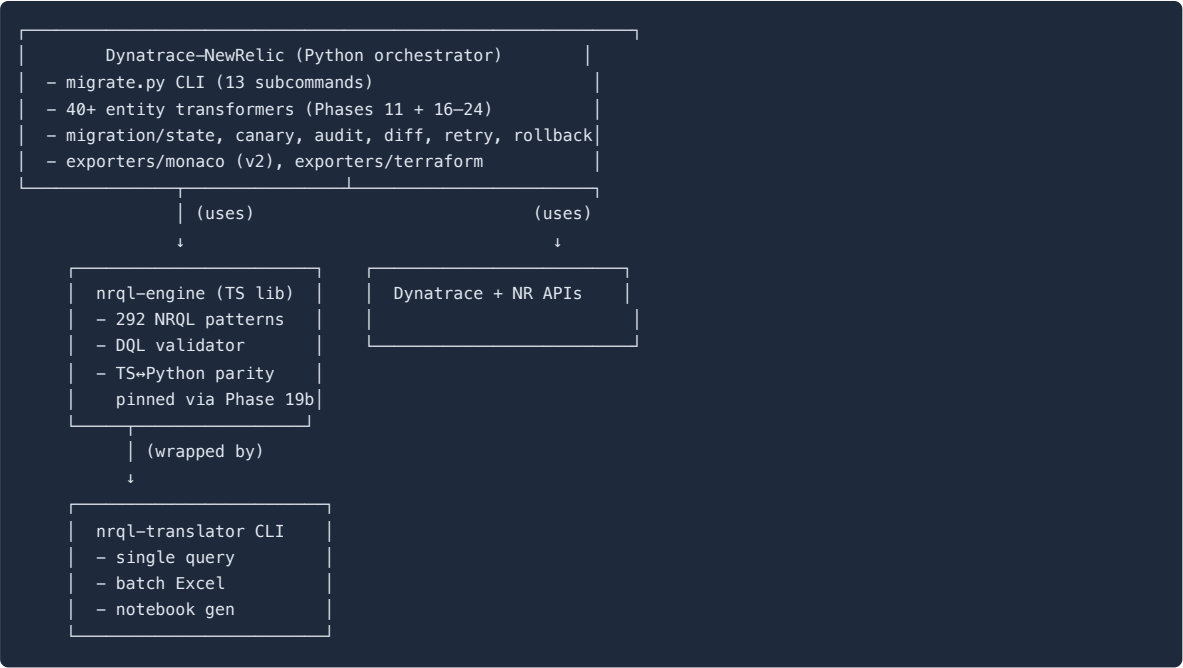
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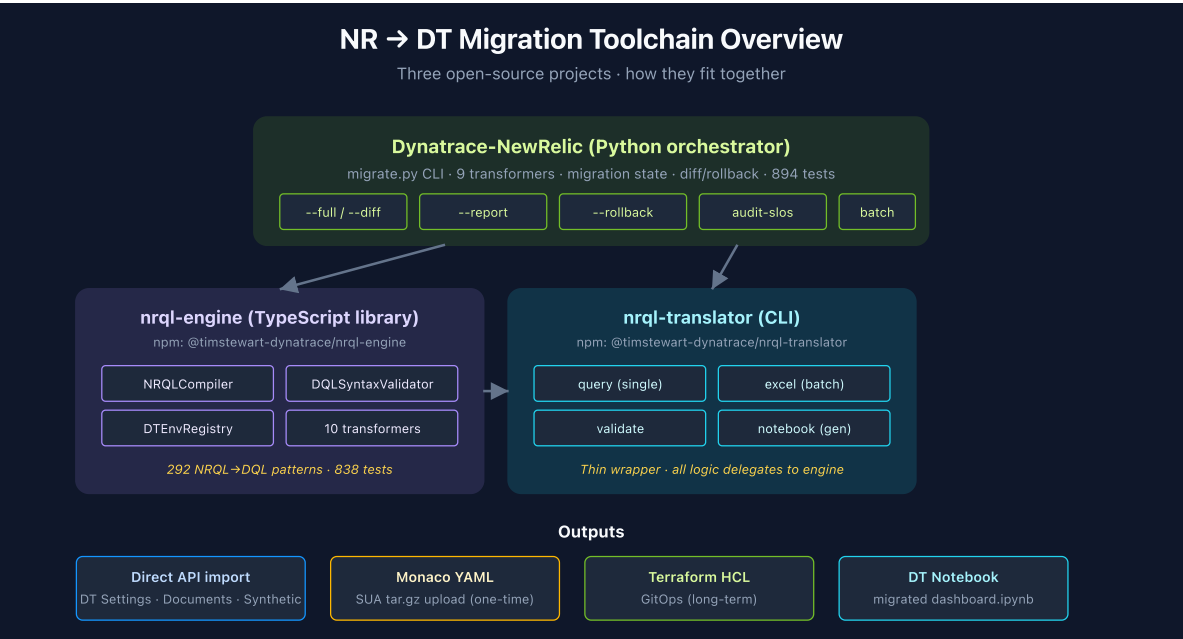
Prerequisites

Requirement	Details
Audience	Engineers operating the migration end-to-end
Required reading	All preceding NRLC notebooks (this is the operational reference)
Access	NR API key + DT API token; both with appropriate scopes (see §2)

1. The Three Projects — How They Fit



Use Case	Tool
End-to-end NR→DT migration	Dynatrace-NewRelic (migrate.py)
Embed translation in your own tool/web app	nrql-engine
One-off NRQL→DQL on the command line	nrql-translator
Translate a batch of dashboard queries	nrql-translator excel or migrate.py batch
Generate a DT notebook from translated queries	nrql-translator notebook



Upstream Home: Dynatrace Migration Assistant (dynatrace-dma)

The official Dynatrace migration tooling lives under the `dynatrace-dma` GitHub organization — the **Dynatrace Migration Assistant** project. Existing repos in the org:

Repo	Source Platform	Status
<code>splunk-to-dynatrace</code>	Splunk	Active
<code>datadog-to-dynatrace</code>	Datadog	Active
<code>newrelic-to-dynatrace</code> (planned)	New Relic	<code>nrql-engine</code> planned to relocate here; pre-staged sweep command in ENGINE-LINKS.md

When the New Relic engine relocates, the recommended install path will become the `dynatrace-dma` repo. Until then, `timstewart-dynatrace/nrql-engine` is canonical.

2. Installation & Configuration

Dynatrace-NewRelic (Python)

```
git clone https://github.com/timstewart-dynatrace/Dynatrace-NewRelic
cd Dynatrace-NewRelic
pip install -r requirements.txt
cp .env.example .env
# Edit .env with NR + DT credentials
```

Future home note: The `nrql-engine` underlying this orchestrator is planned to relocate to the `dynatrace-dma` (**Dynatrace Migration Assistant**) organization, joining the existing `splunk-to-dynatrace` and `datadog-to-dynatrace` repos. Track that org for the official Dynatrace-supported install going forward.

Required NR API Token Scopes

- `read:dashboards` , `read:alerts` , `read:nrql` , `read:synthetics` , `read:slo` , `read:workloads` , `read:logs`

Required DT API Token Scopes

- `dashboards.read` + `dashboards.write` (or Documents v2: `documents:read` , `documents:write`)
- `metrics.read`
- `entities.read`
- `settings:objects.read` + `settings:objects.write`
- `slo.read` + `slo.write`
- `synthetic.read` + `synthetic.write`
- `notifications.write`
- `iam-policies.read` (for IAM policy + OpenPipeline configuration validation)

Note: These are Classic API token scopes (the `<area>.<action>` style). **Platform Tokens use policy-based permissions** instead and are the recommended default on Gen3 tenants — see Dynatrace IAM docs for the equivalent policy statements. OAuth clients follow the Classic scope style for programmatic access.

Configuration File (`.env`)

```
NR_API_KEY=NRAX-...
NR_ACCOUNT_ID=1234567
NR_REGION=US           # or EU

DT_TENANT_URL=https://abc12345.live.dynatrace.com
DT_API_TOKEN=dt0c01.XXXX...

# Optional
MIGRATION_OUTPUT_DIR=./output
DRY_RUN=false
LOG_LEVEL=INFO
```

Settings are loaded via Pydantic; missing required values fail fast with descriptive errors.

3. `migrate.py` CLI Reference

Subcommand inventory (post-Phase-24)

The `migrate.py` entry point registers **13 subcommands**:

`agents`, `archive`, `audit`, `audit-slos`, `batch`, `compile`, `convert`, `export-monaco`, `export-terraform`, `migrate`, `preflight`, `reference`, `scan-instrumentation`

Subcommand	Purpose
<code>migrate</code>	Full pipeline (export → transform → import). Accepts <code>--full</code> , <code>--export-only</code> , <code>--import-only</code> , <code>--dry-run</code> , <code>--components</code> , <code>--rollback <file></code> , <code>--retry <file></code> , <code>--resume</code> , <code>--incremental</code> , <code>--report</code> , <code>--diff</code> , <code>--legacy</code> , <code>--canary <pct></code> , <code>--canary-auto-proceed</code>
<code>compile</code>	Single NRQL → DQL compilation. Flags: <code>--interactive</code> , <code>--file</code> , <code>--validate</code> , <code>--output</code>
<code>convert</code>	NRQL → DQL with post-processing + auto-fix (wraps <code>compile</code> + <code>DQLFixer</code>)
<code>batch</code>	CSV / Excel batch compile (NRQL column)
<code>reference</code>	Print NRQL → DQL reference table; <code>--mappings</code> for full mapping tables (230 metrics + 72 attributes + 90+ aggregations + 34 event types)
<code>preflight</code>	Phase 14. Probe target DT tenant for Gen3 API availability (Settings 2.0 / Document / Automation). Suggests <code>--legacy</code> if any surface is missing.
<code>agents</code>	Phase 16. Per-language APM agent migration action plans (Java / .NET / Node.js / Python / Ruby / PHP / Go). Flags: <code>--language <lang></code> , <code>--phase</code> , <code>--dry-run</code> .
<code>scan-instrumentation</code>	Phase 16. Scan source tree for <code>newrelic.*()</code> SDK calls; emit DT/OTel replacement suggestions (side-effect-free; manual apply).
<code>archive</code>	Phase 17. Pre-decommission NRDB snapshot (resumable JSONL per event type).
<code>audit</code>	Phase 20. Drift detection vs live tenant (also the drift-audit command for baseline-vs-current comparison). Reports RENAMED / DELETED / MODIFIED / EXTRA. Exits 1 on drift.
<code>audit-slos</code>	Validate DT SLOs against live metrics for missing/invalid keys (Phase 11 + Phase 20 enrichment).
<code>export-monaco</code>	Emit Monaco v2 project YAML (Gen3 default; <code>--legacy</code> for Gen2 shapes).
<code>export-terraform</code>	Emit Terraform HCL with <code>dynatrace-oss/dynatrace</code> provider (Gen3 default; <code>--legacy</code> for Gen2 shapes).

Note: Behavioral validation (running NRQL on NR and DQL on DT and diffing results) is handled via the `audit` subcommand using a captured baseline, not a separate `compare` subcommand.

Common flags

Flag	Purpose
<code>--components</code> <code><list></code>	Restrict to specific entity types
<code>--filter</code> <code><pattern></code>	Filter entities by name regex
<code>--output <dir></code>	Override output directory
<code>--log-level</code> <code><level></code>	DEBUG / INFO / WARN / ERROR
<code>--no-color</code>	Disable colored output (CI-friendly)
<code>--legacy</code>	Gen2 output path — emits Alerting Profiles / Management Zones / Auto-Tags / Config v1 dashboards instead of Gen3 equivalents. Always warns at startup.
<code>--canary <pct></code>	Phase 20. Two-wave import — import N% first, await approval, then import remainder
<code>--canary-auto-proceed</code>	Promote canary to full import automatically if drift audit passes
<code>--report</code>	Emit enriched conversion report (confidence_score, warning_codes, runbook_url per entry)

Transformer coverage

`migrate.py` orchestrates 40+ entity transformers across 9 phase batches. See [COVERAGE-MATRIX.md](#) for the complete NR-surface → transformer mapping. High-level summary:

Phase	Transformers added
11 (foundation)	alert, dashboard, synthetic, slo, workload, infrastructure, log_parsing, tag, drop_rule
16 (P0 coverage)	agents/ orchestrators (7 languages), lambda, browser_rum, mobile_rum, custom_instrumentation_translator
17 (P1 alerts + data + identity)	non_nrql_alert, baseline_alert, lookup_table, maintenance_window, change_tracking, custom_event_ingest, identity, log_obfuscation, tools/nrdb_archive
18 (specialized products)	cloud_integration, kubernetes, aiops, vulnerability, npm, ai_monitoring, prometheus
19 (dashboard widget parity)	funnel / honeycomb / event-feed / cascading vars / saved views / permissions uplift
19b (nrql-engine parity)	compiler/shorthands, K8s metric + entity-field overrides, 24 DQL fixer methods
20 (operational safety)	migration/canary, migration/audit, DynatraceClient.delete_entity, enriched ConversionReport
23 (second-wave parity)	key_transaction, otel_metrics, statsd, cloudwatch_metric_streams, metric_transform plugin, transformers/mappings/
24 (third-wave parity)	database_monitoring (10 engines), on_host_integration (12 techs), security_signals, custom_entity, log_archive, metric_normalization, synthetic_specialized (cert-check + broken-links), saved_filter_notebook, otel_collector, legacy/error_inbox_v1, legacy/request_naming_v1

4. Component Selection

Migrating one entity type at a time is the safe pattern. The `--components` flag accepts a comma-separated list:

Component	Migrates	Transformer
dashboards	NR Dashboards → DT Documents (one per page), including funnel / honeycomb / event-feed / cascading variables / saved views	dashboard_transformer (Phase 11 + Phase 19 widget parity)
alerts	NR NRQL conditions → builtin:davis.anomaly-detectors + Workflow	alert_transformer (Phase 11 + Phase 17 extensions)
baseline_alerts	NR baseline / outlier NRQL conditions → Dynatrace Intelligence adaptive detectors	baseline_alert_transformer (Phase 17)
non_nrql_alerts	Infrastructure / Synthetic / Browser / Mobile / External-service / Multi-location-synthetic conditions	non_nrql_alert_transformer (Phase 17)
lookup_tables	NR WHERE IN lookups → Resource Store JSONL + DQL lookup subquery	lookup_table_transformer (Phase 17)
maintenance_windows	NR scheduled + recurring maintenance + mute rules → DT Maintenance Windows + detector filters	maintenance_window_transformer (Phase 17)
change_tracking	NR deployment markers → DT events API (CUSTOM_DEPLOYMENT , CUSTOM_CONFIGURATION)	change_tracking_transformer (Phase 17)
custom_event_ingest	NR custom events → bizevent CloudEvent payloads	custom_event_ingest_transformer (Phase 17)
synthetics	NR Ping / Browser / Scripted API / Scripted Browser → builtin:synthetic_test	synthetic_transformer (Phase 11)
synthetics_specialized	CERT_CHECK + BROKEN_LINKS	synthetic_specialized_transformer (Phase 24)
slos	NR SLOs (v1 / v2) → builtin:monitoring.slo	slo_transformer (Phase 11)
key_transactions	NR Key Transactions → SLO + OpenPipeline enrichment + Workflow bundle	key_transaction_transformer (Phase 23)
workloads	NR Workloads → builtin:segment + bucket-scoped IAM	workload_transformer (Phase 11, Gen3 default)
tags	NR tag rules → OpenPipeline enrichment (builtin:openpipeline.*)	tag_transformer (Phase 11, Gen3 default)
logs	NR log forwarding configs → DT ingest or OneAgent log collection runbook	log_parsing_transformer + runbook (Phase 11)
drops	NR drop rules → OpenPipeline filter processors	drop_rule_transformer (Phase 11)
parsing	NR Grok parsing → OpenPipeline DPL parse processors	log_parsing_transformer (Phase 11)
log_obfuscation	NR PII / PAN masking → OpenPipeline mask processors (7 presets + regex)	log_obfuscation_transformer (Phase 17)
log_archive	NR Log Live Archive → Grail bucket + OpenPipeline egress (S3/GCS/Azure Blob)	log_archive_transformer (Phase 24)
notifications	NR Notification Channels → Workflow action tasks (email/slack/pagerduty/webhook first-class; Jira/ServiceNow/OpsGenie/xMatters/VectorOps/Teams via http-function fallback or --legacy typed shapes)	folded into alert_transformer (Phase 11 + 17)
identity	NR Users / Teams / Roles / SAML → builtin:iam.* + SCIM runbook	identity_transformer (Phase 17)
cloud_integrations	AWS (16 services) / Azure (8 resources) / GCP (8 services)	cloud_integration_transformer (Phase 18)

Component	Migrates	Transformer
kubernetes	NR K8s → DynaKube (full-stack / host-only) + Helm values	kubernetes_transformer (Phase 18)
aiops	NR AI Workflows + enrichments + decisions → DT automation + Dynatrace Intelligence detectors	aiops_transformer (Phase 18)
vulnerability	NR Vuln Mgmt → RVA alerting + per-CVE muting	vulnerability_transformer (Phase 18)
npm	SNMP devices + NetFlow collector (secrets redacted)	npm_transformer (Phase 18)
ai_monitoring	Model registry + inference NRQL→DQL mapping	ai_monitoring_transformer (Phase 18)
prometheus	NR Prometheus → DT scrape + OTLP remote-write + relabel filters	prometheus_transformer (Phase 18)
otel_metrics	NR OTLP metrics ingest → builtin:otel.ingest.metrics + collector YAML	otel_metrics_transformer (Phase 23)
otel_collector	NR OTEL collector (traces + metrics + logs + 5 processor kinds)	otel_collector_transformer (Phase 24)
statsd	NR StatsD → ActiveGate builtin:statsd.metrics + tag-mapping translation	statsd_transformer (Phase 23)
cloudwatch_metric_streams	NR Firehose path → builtin:aws.metric-streams + Terraform snippet	cloudwatch_metric_streams_transformer (Phase 23)
database_monitoring	10 DB engines (MySQL / Postgres / MSSQL / Oracle / MongoDB / Redis / Cassandra / MariaDB / DB2 / HANA) → builtin:dynatrace.extension.db.*	database_monitoring_transformer (Phase 24)
on_host_integrations	12 integrations (NGINX / HAProxy / Kafka / RabbitMQ / Elasticsearch / Memcached / Couchbase / Consul / Apache / etcd / Varnish / Zookeeper) → DT extensions	on_host_integration_transformer (Phase 24)
security_signals	NR Security Signals / IAST → AppSec envelope + per-signature OpenPipeline enrichment	security_signals_transformer (Phase 24)
custom_entity	NR custom entities → DT custom-device POST payload + enrichment matcher	custom_entity_transformer (Phase 24)
metric_normalization	NR rename / aggregate / drop rules → OpenPipeline metric processors	metric_normalization_transformer (Phase 24)
saved_filter_notebook	NR Data Apps → Document API type=='notebook' with markdown + DQL cells	saved_filter_notebook_transformer (Phase 24)

Recommended order: workloads, tags first (foundations), then dashboards, synthetics, then alerts, notifications, baseline_alerts, non_nrql_alerts, maintenance_windows, then slos, key_transactions, then logs, drops, parsing, log_obfuscation, log_archive, then specialized (cloud_integrations, kubernetes, prometheus, database_monitoring, on_host_integrations, vulnerability, security_signals, ai_monitoring, otel_metrics, otel_collector, statsd, cloudwatch_metric_streams, npm, custom_entity, metric_normalization, saved_filter_notebook).

5. Export Formats — Monaco & Terraform

Beyond direct API import, Dynatrace–NewRelic can emit **config-as-code** in two formats:

Monaco YAML

```
python3 migrate.py export-monaco --input ./output --output ./monaco-project
```

Output: a Monaco v2 project (`manifest.yaml` + per-config-type directories with `config.yaml` + JSON templates). Suitable for the SaaS Upgrade Assistant tar.gz upload pattern (see ENBR-90).

Terraform HCL

```
python3 migrate.py export-terraform --input ./output --output ./terraform-project
```

Output: HCL files using the `dynatrace-oss/dynatrace` provider. Resources include `dynatrace_dashboard`, `dynatrace_alerting`, `dynatrace_metric_event`, `dynatrace_slo`, `dynatrace_openpipeline`, etc.

Why both formats

- **Monaco** is best for one-time bootstrap and bulk upload via SUA
- **Terraform** is best for ongoing GitOps-style management

Per ADR-016 in the USFOODS series, **Terraform is the long-term tool**; Monaco is for export and one-time deploys.

6. End-to-End Runbook

This is the operational sequence to run a complete NR→DT migration.

Prerequisites

- [] NR API key + account ID configured
- [] DT Platform Token configured (OAuth / Classic tokens supported but Platform Token is the recommended default)
- [] Bucket strategy in place (USFOODS-G.02)
- [] Host group strategy in place (USFOODS-G.01)
- [] Stakeholders identified per wave

Phase 0 — Preflight (new in Phase 14; always do this first)

```
python3 migrate.py preflight
```

Probes the target tenant for Settings 2.0 / Document API / Automation API availability. If any Gen3 surface is missing, the command suggests `--legacy` and exits non-zero — catch this before burning a run.

Phase 1 — Discover (pair with NR2DT-01)

```
python3 migrate.py migrate --export-only --output ./inventory
python3 migrate.py migrate --report --input ./inventory
```

Review `inventory/exports/newrelic_export.json`, enriched conversion-quality report (`confidence_score`, `warning_codes`, `runbook_url` per entry), and gap analysis. Stakeholder sign-off on wave plan.

Pre-decommission archive (Phase 17):

```
python3 migrate.py archive --event-types Transaction,Log,SyntheticCheck --output ./nrdb-archive
```

Historical NRDB data is **not migratable to Grail** — archive as JSONL before decommissioning NR. Resumable per-event-type cursors let you interrupt and restart safely.

Phase 2 — Translate & Stage

```
# Translate NRQL inventory
# Extract NRQL from the export (produces inventory/all-nrql.txt + .csv)
python3 migrate.py extract-nrql --input ./inventory --output ./inventory/all-nrql.txt
python3 migrate.py extract-nrql --input ./inventory --output ./inventory/all-nrql.csv

# Translate – compile for plain DQL; batch for a CSV with confidence scores
python3 migrate.py compile --file inventory/all-nrql.txt --output translated.dql
python3 migrate.py batch --file inventory/all-nrql.csv --output translated.csv

# Instrumentation scan for newrelic.*() SDK calls (Phase 16)
python3 migrate.py scan-instrumentation --src-root ./app-source --output ./instrumentation-todos.json

# APM agent plan per language (Phase 16)
python3 migrate.py agents --language java --dry-run
python3 migrate.py agents --language nodejs --dry-run

# Transform each component (dry-run)
python3 migrate.py migrate --transform-only --components workloads,tags --dry-run
```

Phase 3 — Wave-by-Wave Import (Gen3 default)

Use `migrate --diff dry-run` and `--canary` for production tenants.

```
# Wave 0 (foundations)
python3 migrate.py migrate --import-only --components workloads,tags --diff
python3 migrate.py migrate --import-only --components workloads,tags --canary 10 --canary-auto-proceed

# Wave 1 (dashboards – read-only impact)
python3 migrate.py migrate --import-only --components dashboards --diff
python3 migrate.py migrate --import-only --components dashboards

# Wave 2 (synthetics, including cert-check + broken-links – Phase 24)
python3 migrate.py migrate --import-only --components synthetics

# Wave 3 (alerts – dual-alert window!)
python3 migrate.py migrate --import-only --components alerts,notifications
# ... wait 1-2 weeks dual-alert ...

# Wave 4 (SLOs + key transactions – Phase 23)
python3 migrate.py migrate --import-only --components slos,key_transactions
python3 migrate.py audit-slos

# Wave 5 (logs/drops/obfuscation/archive – Phase 17 + 24)
python3 migrate.py migrate --import-only --components logs,drops,parsing,log_obfuscation,log_archive

# Wave 6 (specialized – Phase 18 + 24)
python3 migrate.py migrate --import-only --components cloud_integrations,kubernetes,prometheus
python3 migrate.py migrate --import-only --components database_monitoring,on_host_integrations
python3 migrate.py migrate --import-only --components vulnerability,security_signals,ai_monitoring
```

Phase 4 — Validate & Cutover (NRLC-08)

Per-wave validation gates:

```
# Drift audit against captured baseline (Phase 20)
python3 migrate.py audit --baseline ./output

# SLO math-equivalence audit
python3 migrate.py audit-slos

# Behavioral sample validation (drift audit drives the NR-vs-DT comparison via baseline)
python3 migrate.py audit --baseline ./output
```

Sign-off per wave.

Phase 5 — Decommission NR

Disable NR alert routes; archive dashboards; halt NR ingest. Retain rollback manifests for 30+ days.

`DynatraceClient.delete_entity` (Phase 20) now dispatches unified Gen3 deletes — rollback actually executes.

7. CI/CD Integration Patterns

Once the migration is run, ongoing config can live in CI:

GitOps with Terraform

```
# .github/workflows/dt-config.yml
name: dt-config
on: [push]
jobs:
  apply:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: hashicorp/setup-terraform@v3
      - run: terraform init
      - run: terraform plan -out=tfplan
      - run: terraform apply tfplan
    env:
      DT_TOKEN: ${ secrets.DT_TOKEN }
```

Translation in CI

Run the compiler in CI to validate any committed DQL:

```
- run: npx nrql-translator validate --file ./queries.nrql
- run: |
    if [[ $(jq '.confidence' result.json) == "LOW" ]]; then
      echo "::error::LOW-confidence translation - manual review required"
      exit 1
    fi
```

Drift Detection

Schedule a daily drift audit to detect manual DT changes that drifted from the GitOps source of truth — Phase 20 `migrate.py` audit :

```
python3 migrate.py audit --baseline ./output --output drift.json
if [[ $(jq '.driftCount' drift.json) -gt 0 ]]; then
  alert_team "DT config drift detected"
fi
```

Summary

The toolchain is open-source, modular, and CI-friendly. `Dynatrace-NewRelic` orchestrates; `nrql-engine` translates; `nrql-translator` exposes the engine on the CLI. Together they cover discovery, translation, transformation, validation, drift audit, canary promotion, and rollback. Pair them with the bucket strategy (USFOODS-G.02) and host grouping strategy (USFOODS-G.01) for a complete migration foundation.

Series Complete

You've reached the end of the NRLC deep-dive series. For the procedural step-by-step migration walkthrough, see the **NR2DT** series (9 procedural steps + NR2DT-99 summary). For reference, NRLC covers:

#	Topic
01	Platform Comparison
02	NRQL → DQL Translation
03	Dashboard Migration
04	Alert & Workflow Migration
05	Synthetic Monitor Migration
06	SLO & Workload Migration
07	Logs, Tags & Drop Rules
08	Validation, Diff & Rollback
09	Toolchain Reference & End-to-End Runbook

This notebook was AI-generated from community-submitted and publicly available sources, including the open-source [Dynatrace-NewRelic](#), [nrql-engine](#) (planned future home: the [dynatrace-dma](#) Dynatrace Migration Assistant organization), and [nrql-translator](#) projects. This notebook series is not officially supported by Dynatrace or New Relic. Always verify information against the official [Dynatrace documentation](#) and [New Relic documentation](#).